

LM1

三大平价理论

1 Interest Rate Parity (IRP)

Covered Interest Rate Parity

公式: $\frac{F}{S} = \frac{1+r_x}{1+r_y}$ (x/y) 一年内单利, 一年以上复利.

套利: 先判断, 再算 Profit.

结论: 通过签订 Forward 可 Fully Hedge. (外国 = y 国 = 本国)
成立时无套利机会. 套利机制促使 永远成立.

Uncovered Interest Rate Parity

公式: $\frac{E(S_1)}{S_0} = \frac{1+r_x}{1+r_y}$, 无 forward rate, 只有预期 $E(S_1)$

作用: 预测未来 FX 走势.

结论: 成立 $\left\{ \begin{array}{l} \textcircled{1} \text{ 本国: No Risk, Risk 不同, R 相等 } \therefore \text{Risk Neutral!} \\ \textcircled{2} \text{ 外国: FX Risk} \end{array} \right.$
短期不成立, 长期成立

Forward Rate Parity

公式: $\frac{F-S_0}{S_0} = E(\% \Delta S)$, unbiased forecast ($F = E(S_1)$)

结论: Covered & Uncovered 同时成立, Forward Rate Parity 才成立. $\therefore$ 短期不成立, 长期成立.

二者对比

	Covered	VS.	Uncovered
Forward:	✓		✗
FX Risk:	✗		✓
公式:	$\frac{F}{S} = \frac{1+r_x}{1+r_y}$		$\frac{E(S_1)}{S_0} = \frac{1+r_x}{1+r_y}$
	↑ Arbitrage 套利		↑ Carry Trade 套利

$$\Delta\%S = \frac{E(S_1)}{S_0} - 1 = \frac{1+r_x}{1+r_y} - 1 = \frac{r_x - r_y}{1+r_y} \approx r_x - r_y$$

$$\approx r_x - r_y \left\{ \begin{array}{l} > 0 \quad E(S_1) > S_0 \quad Y \uparrow, X \downarrow \\ < 0 \quad E(S_1) < S_0 \quad Y \downarrow, X \uparrow \end{array} \right.$$

2 Purchase Power Parity (PPP) 通胀率与 FX 的关系

Relative PPP

公式: $\frac{E(S_1)}{S_0} = \frac{1+I_x^e}{1+I_y^e}$ $\left\{ \begin{array}{l} \star \text{ Ex-Ante 事前: focus on expected changes } \frac{E(S_1)}{S_0} = \frac{1+I_x^e}{1+I_y^e} \\ \text{Ex-Post 事后: } \frac{S_1}{S_0} = \frac{1+I_x}{1+I_y} \end{array} \right.$

作用: 预测未来 FX 走势 (长期), 通胀高的货币贬值.

结论: 短期不成立, 长期成立

$$\Delta\%S = \frac{E(S_1) - S_0}{S_0} = \frac{E(S_1)}{S_0} - 1 = \frac{1+I_x^e}{1+I_y^e} - 1 = \frac{I_x^e - I_y^e}{1+I_y^e} \approx I_x^e - I_y^e$$

$$\approx I_x^e - I_y^e \left\{ \begin{array}{l} > 0 \quad I_x^e > I_y^e, X \downarrow, Y \uparrow \\ < 0 \quad I_x^e < I_y^e, X \uparrow, Y \downarrow \end{array} \right.$$

Absolute PPP (基于 Law of one price)

3 International Fisher Effect

假设: Real Interest Rate Parity, $R_x^* = R_y^*$, 各国真实利率水平相等. $\left\{ \begin{array}{l} X: (1+R_N^*) = (1+R_x^*) \cdot (1+I_x^e) \\ Y: (1+R_N^*) = (1+R_y^*) \cdot (1+I_y^e) \end{array} \right.$

公式: $\frac{1+r_x}{1+r_y} = \frac{1+I_x^e}{1+I_y^e}$ 变形 $\rightarrow r_x - r_y = I_x^e - I_y^e$

结论: 名义利率差与通胀预期差相等
短期不成立, 长期成立

三者关系 (x/y)

Uncovered R & FX	$\frac{E(S_1)}{S_0} = \frac{1+r_x}{1+r_y}$
International Fisher r & I	$\frac{1+r_x}{1+r_y} = \frac{1+I_x^e}{1+I_y^e}$
Ex Ante Relative PPP I & FX	$\frac{E(S_1)}{S_0} = \frac{1+I_x^e}{1+I_y^e}$

两个成立时, 第三个自然成立

$$\text{均成立时, } \Delta\%S = \frac{E(S_1) - S_0}{S_0}$$

$$\approx r_x - r_y$$

$$\approx I_x - I_y$$

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Bid-Ask Spread



$$\text{Spread} = \text{Ask (offer) Price} - \text{Bid Price} \quad 1 \text{ pip} = 0.0001$$

Interbank Spread 关注 Risk-流动性, 作为 spread benchmark.

Triangular Arbitrage 三角套利

- ① Cross Rate 计算: 计算先判断 $\times / \div$, 相乘同边, 相除对边.
- ② 判断套利流程: ① & ② 市与 ③ 市相比, 买低卖高. 决定先在 ①② 换便宜币.
- ③ 套利 Profit: 乘小除大

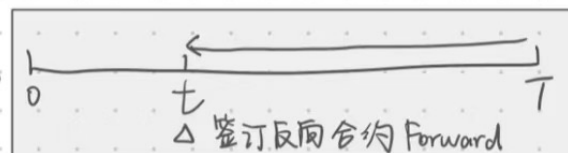
$$\text{Forward Point} = F - S \quad \begin{cases} > 0, F > S, \text{Premium 升水} \\ < 0, F < S, \text{Discount 贴水} \end{cases}$$

Mark to Market Value

① 先画图. 签订反向合约.

② 计算 $V_T = \text{收} - \text{支}$

③ 去年化 $V_t = V_T \text{折现} = \frac{V_T}{1 + r^{T-t}}$ 单利形式



FX Carry Trade 套息交易

计算: $A/B, r_A > r_B \rightarrow$ B 国借, A 国投

$$B \xrightarrow[t=0]{S_0} A \xrightarrow[t=1]{\frac{1+r_A}{S_0 \times (1+r_A)}} A' \xrightarrow[t=1]{\frac{S_1}{S_0} \times (1+r_A)} B'$$

$$\text{Profit} = \text{收} - \text{支} = \frac{S_0}{S_1} (1+r_A) - (1+r_B) = r_A - r_B - \Delta\% S$$

$\left\{ \begin{array}{l} A \downarrow, B \uparrow \text{ 导致 } S_1 > S_0, \Delta\% S > 0, \text{Profit} \downarrow \\ A \uparrow, B \downarrow \text{ 导致 } S_1 < S_0, \Delta\% S < 0, \text{Profit} \uparrow \end{array} \right.$

结论: 交易基于利率差, 交易大额

当 Uncovered IRP 成立时, $\frac{S_1}{S_0} = \frac{1+r_A}{1+r_B}$, 套息交易 Profit = 0.

而 Uncovered IRP 短期不成立, 长期成立, $\therefore$ 短期仍可获利.

Risk: Investment Currency 贬值, Financing Currency 升值.

Crash Risk: Leverage 导致崩盘风险. volatility 高时不建议 trade. 概率 $\sim$ 高峰肥尾.

Balance of Payment (BOP 账户 & FX 关系).

- Current Account 经营性账户: 国际贸易, 本币贬值改善贸易赤字.
- Financial Account 资本账户: 资本流通, 对 FX 影响更大.

$r_{RMB} \uparrow \rightarrow \text{资本流入中国} \rightarrow \text{兑换 RMB} \rightarrow D_{RMB} \uparrow \rightarrow r_{RMB} \downarrow$

短期 FX 影响因素: $r, I, \text{risk premium, long-run fair value.}$

FX 管理.

Emerging market policymaker have greater success. (有效性).

Warning Signs of Currency Crisis

- 1 造成本币深度贬值的行為.
- 1 负面信息.

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Exchange Rate Determination 汇率决定模型

★ Mundell-Pleming Model 研究短期货币财政政策影响

1 Floating high 浮动汇率下资本高度流通 <F.A.>

货币政策: 扩张 $\rightarrow r \downarrow \rightarrow$ 资本流出 $\rightarrow$ 本币 $\downarrow$

<F.A.>

$\Rightarrow$ 本币 $\downarrow$

扩张 $\rightarrow$ 消费 $C \uparrow \rightarrow$ 进口 $M \uparrow \rightarrow$ 逆差 $\rightarrow$ 本币 $\downarrow$

<C.A.>

财政政策: 扩张 $\rightarrow$ 政府支出 $G \uparrow \rightarrow$ 赤字 $D \uparrow \rightarrow r \uparrow \rightarrow$ 资本流入 $\rightarrow$ 本币 $\uparrow$

<F.A.>

$\Rightarrow$ 本币 $\uparrow$ (F.A. 起决定作用)

扩张 $\rightarrow$ 消费 $C \uparrow \rightarrow$ 进口 $M \uparrow \rightarrow$ 逆差 $\rightarrow$ 本币 $\downarrow$

<C.A.>

2 Floating low <C.A.>

货币政策: 扩张 $\rightarrow r \downarrow \rightarrow$ 资本流出 $\rightarrow$ 本币 $\downarrow$

<F.A.>

$\Rightarrow$ 本币 $\downarrow$

扩张 $\rightarrow$ 消费 $C \uparrow \rightarrow$ 进口 $M \uparrow \rightarrow$ 逆差 $\rightarrow$ 本币 $\downarrow$

<C.A.>

财政政策: 扩张 $\rightarrow$ 政府支出 $G \uparrow \rightarrow$ 赤字 $D \uparrow \rightarrow r \uparrow \rightarrow$ 资本流入 $\rightarrow$ 本币 $\uparrow$

<F.A.>

$\Rightarrow$ 本币 $\downarrow$ (C.A. 起决定作用)

扩张 $\rightarrow$ 消费 $C \uparrow \rightarrow$ 进口 $M \uparrow \rightarrow$ 逆差 $\rightarrow$ 本币 $\downarrow$

<C.A.>

3 Fixed high <F.A.>

货币政策: 紧缩 $\rightarrow r \uparrow \rightarrow$ 资本流入 $\rightarrow$ 本币 $\uparrow$ 固定汇率 $\rightarrow$ 抽本, 扩张 政策失效

财政政策: 扩张 $\rightarrow$ 政府支出 $G \uparrow \rightarrow$ 赤字 $D \uparrow \rightarrow r \uparrow \rightarrow$ 资本流入 $\rightarrow$ 本币 $\uparrow$ 固定汇率 $\rightarrow$ 抽本, 扩张

三元悖论

1. 固定FX
2. 资本自由流通
3. 独立货币政策

Monetary Model 研究货币政策

货币主义思想: $MV = PY$ $M_S \uparrow \rightarrow P \uparrow \rightarrow I \uparrow \rightarrow$ 本币 $\downarrow \rightarrow$ FXN变化, FXR不变

pure M.M. 长期

Dornbusch overshooting $\begin{cases} \text{短期} & \text{M-P Floating high } <F.A.> & \text{FXR变} \\ \text{长期} & \text{pure M.M.} & \text{FXN变, FXR不变} \end{cases} \Rightarrow \text{FXR} \rightarrow \text{均值回归}$

Portfolio Balance Model 研究长期下财政政策影响

财政: 扩张 $\rightarrow G \uparrow T \downarrow \rightarrow$ 赤字 (失去信心) $\rightarrow$ 资本流出 $\rightarrow$ 本币 $\downarrow$ 与M-F相反

掌握 M.M. PBM 与 MFM 的对比

宽松财政: 短期: M-F 本币 $\uparrow$
长期: PBM 本币 $\downarrow$

LN2. Economic Growth

G-K Model 上市公司分解 Why GDP Growth matters to investors?

$$ECR = \text{dividend yield} + \Delta(P/E) + \underbrace{v + g + \Delta S}_{\Delta S = \text{net buybacks} + \text{relative dynamism of economy}}$$

$\Delta S = \text{net buybacks} + \text{relative dynamism of economy}$

Potential GDP. (资源全部利用上的最大产出)

1 股市与 Potential GDP 增长率的关系

$$P = \text{GDP} \times \frac{E}{\text{GDP}} \times \frac{P}{E}$$

$$\ln P = \ln \text{GDP} + \ln E/\text{GDP} + \ln P/E$$

$$\frac{\Delta P}{P} = \frac{\Delta \text{GDP}}{\text{GDP}} + \frac{\Delta E/\text{GDP}}{E/\text{GDP}} + \frac{\Delta P/E}{P/E}$$

$$\Delta\%P = \Delta\%\text{GDP} + \Delta\%E/\text{GDP} + \Delta\%P/E$$

结论: 长期 $\begin{cases} \Delta\%(E/\text{GDP}) = 0 \\ \Delta\%(P/E) = 0 \end{cases} \Rightarrow \Delta\%P = \Delta\%\text{GDP}$

2 GDP持续增长要伴随 high real interest rate

3 High potential GDP improves general credit quality of 债券

★ Production Function

1 辨析: $Y = AK^\alpha L^{1-\alpha}$ 规模报酬不变 (前提是 K, L 贡献度之和为 1)

2 变形

① 人均 y

公式: $y = AK^\alpha$

结论: 1 k 边际递减



$\begin{cases} \alpha < 1, \text{ 边际递减} \\ \alpha > 1, \text{ 边际递增} \end{cases}$

2 y 增加依靠

Capital Deepening 资本深化 k ↑

Technological Progress 技术进步 A ↑, y 持续增长依靠 技术进步

3 发展中国家资本深化效果更好

② $\Delta\%Y$, The growth accounting equation

$$\Delta\%Y = \Delta\%A + \alpha\Delta\%K + (1-\alpha)\Delta\%L$$

③ $L \rightarrow Y$, Labor productivity growth accounting equation

$$Y = y \times L \Rightarrow \Delta\%Y = \Delta\%y + \Delta\%L$$

Factors

1 Natural Resources

无直接关系

Resource Curse

Fail to develop economic institutions

Dutch Disease: 石油出口 ↑ → 顺差 → 本币 ↑ → 其他行业出口 ↓

2 Labor Supply

population growth

Labor Force Participation = Labor Force / Working Age Population

Net Migration 净移民

Average hours worked

Labor Quality

3 Capital

ICT 高科技, A ↑ / Non ICT 非高科技, K ↑ (physical capital)

Public Infrastructure

LM2

Theories

1 Classical Growth Theory 古典. 外生. (不考).

结论: 技术(偶然) $\uparrow \rightarrow GDP \uparrow \rightarrow$ 人口 $\uparrow \rightarrow$ 人均GDP $\downarrow \rightarrow$ 稳定(暂时).
稳定在 Subsistence real wage, 支持生活的实际工资底线.

2 Neoclassical Growth Theory 新古典. 外生. (技术进步不可控)

长期 E^* Assumption $\left\{ \begin{array}{l} \frac{Y}{K} \text{ 稳定} \\ \Delta\%K = \Delta\%Y \end{array} \right. \Rightarrow \Delta\%Y = \frac{\Delta\%A}{1-\alpha} \Rightarrow g^* = \frac{\theta}{1-\alpha}$
 $G^* = \frac{\theta}{F\alpha} + \Delta\%L$

结论: 技术(偶然) $\uparrow \rightarrow K \uparrow \rightarrow K \uparrow$ 边际递减 $Y \uparrow \rightarrow$ 稳定.

趋同.

3 Endogenous Growth Theory 内生 (R&D $\uparrow \rightarrow$ 技术持续进步)

结论: 不趋同于 g^*, G^* , 增长是长期的.

v.s. 新古典: 新古典: 新技术(偶然) $\uparrow \rightarrow K \uparrow \rightarrow Y \uparrow$ 短暂, 非长期持续.
内生: 技术 $\uparrow \rightleftharpoons K \uparrow \rightarrow Y \uparrow$ 长期持续. 不存在边际递减. 稳定, 趋同...

Convergence.

- Absolute convergence 假设技术流动无限制; $\Delta Y/Y$ 会趋同, Y 不会趋同.
- Conditional Convergence same saving rate, pop. growth rate, production function; Y, G^*, g^* 会趋同.
- Club convergence. Only rich and middle-income countries Y 趋同. poor country 须要政治改革

LM3 监管

Rationale for Regulation

- 1 Information Friction.
 - Lack of access
 - Asymmetrical Information
- 2 Externalities 外部性 - negative externality:
 - systemic risk
 - financial contagion
- 3 Weak Competition

Classifications:

- 1 Regulations
 - Statutes: broad level
 - Administrative: detailed level
 - Judicial law 司法
- 2 Regulators
 - Government agencies
 - Outside bodies: 制度规定, 不检查
 - Independent regulators: 有执法权, 但非政府机构, 公立机构, 不依靠政府资助
 - Industry self-regulatory bodies: regulate members, not have the force of law, private.
 - SRO: 政府授权, 有执法权
 - non-SRO: 无政府授权, 无执法权

	Independent Regulator	SROs
同:	① Less political pressure ② Not funded by government ③ Authorized by government.	
不同:	public all	private member

3 Regulatory Interdependencies

- Regulatory capture theory 监管者被捕获
- Regulatory competition: 主动提供宽松环境以提高吸引力
- Regulatory Arbitrage.

4 Tools

Bail in tools: 失败机构股东、债主承担损失, 而非纳税人

Cost-Benefit Analysis of Regulation